

Step 4: Test replication	45
Step 5: Clean up AWS DMS resources	48
AWS DMS Fleet Advisor end of support	50
AWS DMS Fleet Advisor end of support details	50
End of support plan	50
FAQ	50
Discovering databases for migration	52
Supported AWS Regions	53
Understand DMS Fleet Advisor	55
Creating required resources	56
Configure resources using CloudFormation	56
Configure Amazon S3 and IAM in the AWS Management Console	59
Creating database users	66
Using a database user	66
Creating a database user with MySQL	67
Creating a database user with Oracle	68
Creating a database user with PostgreSQL	68
Creating a database user with SQL Server	69
Deleting database users	70
Data collectors	73
Permissions	74
Creating a data collector	74
Installing a data collector	77
Discovering OS and database servers	80
Managing monitored objects	84
Using SSL	86
Collecting data	88
Troubleshooting	93
Inventory	97
Using the database inventory	98
Using the schema inventory	100
Target recommendations	101
Target instances	102
How does DMS Fleet Advisor determine target specifications?	103
Generating target recommendations	104
Recommendation details	106

Exporting target recommendations	108
Migration limitations	109
Troubleshooting	126
Limitations	128
Converting database schemas	129
How it works	130
Key capabilities	131
Supported conversion paths	132
Supported AWS Regions	133
Getting started	135
Prerequisites	135
Create an instance profile	140
Create data providers	140
Create a migration project	144
Create an assessment report	145
Convert your source code	146
Apply the converted code	146
Clean up and troubleshoot	147
Concepts	148
Metadata model	148
Selection rules	157
Transformation rules	179
Setting up a network	209
Single VPC configuration	210
Multiple VPC configuration	211
Shared VPC configuration	213
Using Direct Connect or a VPN	214
Using an internet connection	215
Resolving domain endpoints using DNS	215
Troubleshooting network issues	216
Creating source data providers	217
Using SQL Server as a source	218
Using Oracle as a source	220
Using Oracle Data Warehouse as a source	221
Using PostgreSQL as a source	221
Using MySQL as a source	222

Using Db2 for Linux, UNIX and Windows as a source	222
Using Db2 for z/OS as a source	224
Using a SAP ASE (Sybase ASE) database	225
Creating and setting target data providers	226
Using MySQL as a target	227
Using PostgreSQL as a target	228
Using Amazon Redshift as a target	229
Using Amazon RDS for Db2 database as a target	230
Virtual mode for offline source and virtual target	231
Create a data provider in Virtual Mode	231
Work with a data provider in Virtual Mode	231
Offline source	232
Virtual target	242
Managing migration projects	244
Specifying migration project settings	245
Access logs for AWS DMS Schema Conversion	245
Conversion assessment reports	246
Running an assessment	246
Saving assessment reports	248
Viewing your assessment report	251
Understanding report metrics	256
Migration planning	260
Schema conversion	264
Converting your database schema: step-by-step guide	265
Converting database objects with generative AI	269
Specifying schema conversion settings	297
Refreshing your database schemas	341
Saving and applying your schema	342
Converting embedded SQL in Java applications	344
AI agents	344
Prerequisites	345
Installing the skill	345
Best practices	347
Examples	348
Using extension packs	351
Homogeneous data migrations	353

Supported AWS Regions	354
Features	356
Limitations	356
Overview	357
Setting up	358
Creating IAM resources	358
Setting up a network	361
VPC peering network configurations	364
Creating source data providers	368
Using MySQL or MariaDB as a source	368
Using PostgreSQL as a source	372
Using MongoDB or Amazon DocumentDB as a source	375
Create and set a target database	379
Using MySQL or MariaDB as a target	379
Using PostgreSQL as a target	381
Using Amazon DocumentDB as a target	382
Migrating data	383
Creating a data migration	383
Selection rules	386
Managing data migrations	388
Monitoring data migrations	390
Migration statuses	392
Migrating data from MySQL	393
Migrating data from PostgreSQL	394
Migrating data from MongoDB	397
Target table preparation mode	398
Troubleshooting	399
Create a data migration	399
Start a data migration	400
Connection issues	400
Views migrate as tables in PostgreSQL	401
Working with migration projects	402
Creating a subnet group	403
Creating instance profiles	403
Creating data providers	405
Creating migration projects	407

Managing migration projects	409
AWS zero-ETL integration for self-managed database sources	410
How zero-ETL integration works for self-managed database sources	410
Source database configuration before creating integration	411
Setting up IAM permissions and encryption for zero-ETL integration	412
Prerequisites	412
Creating a KMS key	413
Configuring the KMS key policy	414
Creating an IAM user and configuring AWS CLI	415
Creating and attaching an IAM policy	416
Configuring the AWS CLI profile	418
Creating a Redshift resource policy	419
Example: Creating a zero-ETL integration with encryption	420
Security best practices	421
Best practices	423
Migration planning for AWS Database Migration Service	423
Schema conversion	425
Reviewing AWS DMS documentation	425
Running a proof of concept	425
Improving performance	426
Using your own on-premises name server	430
Using Amazon Route 53 Resolver with AWS DMS	431
Migrating large binary objects (LOBs)	432
Using limited LOB mode	433
Improved LOB performance	434
Improving performance when migrating large tables using row filtering	437
Ongoing replication	438
Reducing load on your source database	439
Reducing bottlenecks on your target database	439
Using data validation	440
Metrics monitoring	440
Events	441
Using the task log	441
Troubleshooting replication with Time Travel	442
Changing the user and schema for an Oracle target	442
Changing table and index tablespaces for an Oracle target	443

Upgrading a replication instance	444
Understanding migration cost	444
Working with AWS DMS Serverless	445
DMS Serverless components	446
Supported Endpoints	449
Creating a serverless replication	450
Modifying AWS DMS serverless replications	453
Compute Config	456
Understanding autoscaling in AWS DMS serverless	457
Monitoring AWS DMS serverless replications	459
Extended Full-Load Throughput Amazon Redshift	463
Understanding Storage Autoscaling in AWS DMS Serverless	464
Serverless limitations	465
Serverless premigrations	466
Using KMS key to encrypt files	467
Limitations	467
Working with replication instances	468
Choosing replication instance types	473
Deciding what instance class to use	478
Unlimited mode burstable instances	479
Sizing a replication instance	480
Factors to consider	481
Common issues	482
Best practices	482
Replication engine versions	483
Upgrading the engine version using the console	483
Upgrading the engine version using the AWS CLI	484
Public and private replication instances	485
IP addressing and network types	485
Setting up a network for a replication instance	487
Network configurations for database migration	488
Creating a replication subnet group	498
Resolving domain endpoints using DNS	499
Setting an encryption key	499
Creating a replication instance	500
Modifying a replication instance	505

Best practices when modifying a replication instance	510
Rebooting a replication instance	511
Deleting a replication instance	514
DMS maintenance window	516
Effect of maintenance on existing migration tasks	516
Changing the maintenance window setting	516
Endpoints	519
Creating source and target endpoints	519
Using IAM authentication for Amazon RDS endpoint in AWS DMS	525
Sources for data migration	529
Using Oracle as a source	530
Using SQL Server as a source	598
Using Azure SQL database as a source	652
Using Azure SQL Managed Instance as a source	652
Using Azure Database for PostgreSQL as a source	652
Using Azure Database for MySQL as a source	654
Using OCI MySQL Heatwave as a source	654
Using Google Cloud for MySQL as a source	655
Using Google Cloud for PostgreSQL as a source	656
Using PostgreSQL as a source	657
Using MySQL as a source	698
Using SAP ASE as a source	715
Using MongoDB as a source	723
Using Amazon DocumentDB as a source	742
Using Amazon S3 as a source	759
Using IBM Db2 LUW as a source	774
Using IBM Db2 for z/OS as a source	784
Targets for data migration	824
Using Oracle as a target	826
Using SQL Server as a target	836
Using PostgreSQL as a target	842
Using MySQL as a target	855
Using Amazon Redshift as a target	865
Using SAP ASE as a target	891
Using Amazon S3 as a target	893
Using Amazon DynamoDB as a target	944

Using Amazon Kinesis Data Streams as a target	966
Using Apache Kafka as a target	985
Using OpenSearch as a target	1013
Using Amazon DocumentDB as a target	1020
Using Amazon Neptune as a target	1033
Using Redis OSS as a target	1050
Using Babelfish as a target	1057
Using Amazon Timestream as a target	1065
Using Db2 as a target	1076
VPC endpoints for data migration	1078
Common AWS DMS prerequisites	1079
Set up an Amazon VPC endpoint with AWS Secrets Manager	1079
Set up an Amazon VPC endpoint with Amazon S3	1081
Setup an Amazon VPC endpoint for Amazon DynamoDB	1083
Setup an Amazon VPC endpoint for Amazon Kinesis	1083
Setup an Amazon VPC endpoint for Amazon Redshift	1084
Setup an Amazon VPC endpoint for Amazon OpenSearch Service	1084
DMS replication instance (provisioned and Serverless), subnet groups, and endpoints	1084
Who is impacted when migrating to AWS DMS versions 3.4.7 and higher?	1088
Who is not impacted when migrating to AWS DMS versions 3.4.7 and higher?	1088
Preparing a migration to AWS DMS versions 3.4.7 and higher	1088
Supported DDL statements	1089
Limitations and considerations	1090
Advanced endpoint configuration	1090
VPC peering configuration for AWS DMS.	1090
Security group configuration for AWS DMS	1093
Network Access Control List (NACL) configuration for AWS DMS	1095
Configuring AWS DMS secrets manager VPC Endpoint	1097
Additional considerations	1099
Tasks	1101
Creating a task	1105
Task settings	1112
Setting LOB support	1168
Creating multiple tasks	1171
Continuous replication tasks	1172
Replication starting from a CDC start point	1174

Performing bidirectional replication	1181
Modifying a task	1185
Moving a task	1185
Reloading tables during a task	1186
AWS Management Console	1187
Table mapping	1188
Specifying table selection and transformations rules from the console	1189
Specifying table selection and transformations rules using JSON	1193
Selection rules and actions	1195
Wildcards in table mapping	1203
Transformation rules and actions	1204
Using transformation rule expressions to define column content	1228
Table and collection settings rules and operations	1244
Using data masking to hide sensitive information	1275
Using source filters	1280
Applying filters	1281
Filtering by time and date	1287
Enabling and working with premigration assessments	1288
Prerequisites	1289
Specifying, starting, and viewing assessment runs	1292
Individual assessments	1296
Starting and viewing data type assessments	1374
Troubleshooting assessment runs	1378
Specifying supplemental data	1380
Monitoring tasks	1381
Task status	1383
Table state during tasks	1386
Monitoring replication tasks using Amazon CloudWatch	1387
AWS Database Migration Service metrics	1388
Replication instance metrics	1391
Replication task metrics	1394
Viewing and managing AWS DMS logs	1398
Logging AWS DMS API calls with AWS CloudTrail	1400
AWS DMS information in CloudTrail	1400
Understanding AWS DMS log file entries	1401
Context logging	1404

Object Types	1405
Logging Examples	1406
Limitations	1407
Enhanced monitoring dashboard	1408
Overview	1408
Viewing metrics in the enhanced monitoring dashboard	1409
Enhanced monitoring dashboard views	1409
Retention of enhanced monitoring metrics	1413
Viewing AWS DMS events	1413
Working with EventBridge events	1417
Using Amazon EventBridge event rules for AWS DMS	1418
AWS DMS event categories and event messages	1419
ReplicationInstance event messages	1419
ReplicationTask event messages	1423
Replication event messages	1425
Working with Amazon SNS events	1427
Moving event subscriptions to Amazon EventBridge	1427
Working with Amazon SNS events and notifications	1428
AWS DMS event categories and event messages for SNS notifications	1430
Subscribing to AWS DMS event notification using SNS	1434
Using the AWS Management Console	1435
Validating the access policy of your SNS topic	1437
Data validation	1439
Replication task statistics	1440
Replication task statistics with Amazon CloudWatch	1443
Revalidating tables during a task	1444
AWS Management Console	1444
Using JSON editor to modify validation rules	1444
Validation only tasks	1445
Full load validation only	1446
CDC validation only	1446
Validation only use cases	1446
Troubleshooting	1447
awsdms_validation_failures_v2 control table	1449
Redshift Validation Performance	1450
Enhanced data validation for AWS Database Migration Service	1451

Prerequisites	1451
Enhanced data validation limitations	1452
Limitations	1452
S3 Validation	1454
Prerequisites	1454
Permissions	1455
Limitations	1456
Validation only tasks	1457
AWS DMS data resync	1458
Limitations	1459
Scheduling and timing	1459
Use cases	1460
Best practices	1461
Data resync configuration and examples	1462
Validation and troubleshooting	1464
Cron expression rules	1466
Tagging resources	1468
API	1469
Security	1472
Data protection	1474
Data encryption	1474
Internetwork traffic privacy	1475
Data protection in DMS Fleet Advisor	1475
Opting out of storing data	1476
Cross-region inference	1477
Identity and access management	1482
Audience	1482
Authenticating with identities	1483
Managing access using policies	1484
How AWS Database Migration Service works with IAM	1485
Identity-based policy examples	1492
Resource-based policy examples	1500
Using secrets to access resources	1505
Using service-linked roles	1515
Troubleshooting	1521
IAM permissions required	1524

IAM roles for the AWS DMS console, CLI and API	1529
Cross-service confused deputy prevention	1535
AWS managed policies	1539
Compliance validation	1552
Resilience	1554
Infrastructure security	1555
Fine-grained access control	1558
Using resource names to control access	1558
Using tags to control access	1561
Encryption for migrations	1568
Encryption for DMS SC	1570
Setting an encryption key	1572
Network security	1575
Using SSL	1577
Limitations on using SSL with AWS DMS	1580
Managing certificates	1580
Enabling SSL for a MySQL-compatible, PostgreSQL, or SQL Server endpoint	1581
Changing the database password	1584
Using Kerberos Authentication	1584
AWS DMS Kerberos Authentication Architecture Overview	1585
Limitations on using Kerberos authentication with AWS DMS	1585
Prerequisites	1586
Enabling Kerberos support on an AWS DMS replication instance	1588
Enabling Kerberos support on a source endpoint	1589
Testing a source endpoint	1590
Limits	1591
Resource quotas for AWS Database Migration Service	1591
Understanding API request throttling	1593
Troubleshooting and diagnostic support	1594
Migration tasks run slowly	1595
Task status bar doesn't move	1596
Task completes but nothing was migrated	1596
Foreign keys and secondary indexes are missing	1596
AWS DMS does not create CloudWatch logs	1597
Issues occur with connecting to Amazon RDS	1597
Error message: Incorrect thread connection string: Incorrect thread value 0	1598

Networking issues occur	1598
CDC is stuck after full load	1599
Primary key violation errors occur when you restart a task	1599
Initial load of a schema fails	1600
Tasks fail with an unknown error	1600
Task restart loads tables from the beginning	1600
Number of tables per task causes issues	1600
Tasks fail when a primary key is created on a LOB column	1600
Duplicate records occur on a target table without a primary key	1601
Source endpoints fall in the reserved IP range	1601
Timestamps are garbled in Amazon Athena queries	1601
Troubleshooting issues with Oracle	1602
Pulling data from views	1602
Migrating LOBs from Oracle 12c	1602
Switching between Oracle LogMiner and Binary Reader	1603
Error: Oracle CDC stopped 122301 oracle CDC maximum retry counter exceeded.	1603
Automatically add supplemental logging to an Oracle source endpoint	1604
LOB changes aren't being captured	1604
Error: ORA-12899: Value too large for column <i>column-name</i>	1605
NUMBER data type being misinterpreted	1605
Records missing during full load	1605
Table Error	1606
Error: Cannot retrieve Oracle archived Redo log destination ids	1606
Evaluating read performance of Oracle redo or archive logs	1606
Failed to get LOB data	1608
Troubleshooting issues with MySQL	1609
CDC task failing for Amazon RDS DB instance endpoint because binary logging disabled	1610
Connections to a target MySQL instance are disconnected during a task	1610
Adding autocommit to a MySQL-compatible endpoint	1611
Disable foreign keys on a target MySQL-compatible endpoint	1611
Characters replaced with question mark	1612
"Bad event" log entries	1612
Change data capture with MySQL 5.5	1612
Increasing binary log retention for Amazon RDS DB instances	1612
Log message: Some changes from the source database had no impact when applied to the target database.	1612

Error: Identifier too long	1613
Error: Unsupported character set causes field data conversion to fail	1613
Error: Codepage 1252 to UTF8 [120112] a field data conversion failed	1614
Indexes, Foreign Keys, or Cascade Updates or Deletes Not Migrated	1614
Troubleshooting issues with PostgreSQL	1616
JSON data types being truncated	1617
Columns of a user-defined data type not being migrated correctly	1617
Error: No schema has been selected to create in	1617
Deletes and updates to a table aren't being replicated using CDC	1618
Truncate statements aren't being propagated	1618
Preventing PostgreSQL from capturing DDL	1618
Selecting the schema where database objects for capturing DDL are created	1618
Oracle tables missing after migrating to PostgreSQL	1618
ReplicationSlotDiskUsage increases and restart_lsn stops moving forward during long transactions, such as ETL workloads	1619
Task using view as a source has no rows copied	1619
Invalid byte sequence for encoding "UTF8"	1619
Troubleshooting issues with Microsoft SQL Server	1620
Ongoing replication fails after RDS for SQL Server fails over to secondary	1620
Errors capturing changes for SQL server database	1621
Missing identity columns	1621
Error: SQL Server doesn't support publications	1621
Changes do not appear in your target	1622
Non-uniform table mapped across partitions	1622
Error : CDC task failed with a Bad Envelope, invalid data context/LCX code while processing a transaction	1623
Troubleshooting issues with Amazon Redshift	1623
Loading in to an Amazon Redshift cluster in a different AWS Region	1624
Error: Relation "awsdms_apply_exceptions" already exists	1624
Errors with tables whose name begins with "awsdms_changes"	1624
Seeing tables in clusters with names like dms.awsdms_changes000000000XXXX	1624
Permissions required to work with Amazon Redshift	1624
Error: column "TASK_NAME" of relation "awsdms_apply_exceptions" does not exist	1625
Troubleshooting issues with Amazon Aurora MySQL	1625
Error: CHARACTER SET UTF8 fields terminated by ',' enclosed by '"' lines terminated by '\n'	1626

Troubleshooting issues with SAP ASE	1626
Error: LOB columns have NULL values when source has a composite unique index with NULL values	1626
Troubleshooting issues with IBM Db2	1627
Error: Resume from timestamp is not supported Task	1627
Table suspended a table with error "Failed to build 'where' statement"	1627
Troubleshooting latency	1628
Types of CDC latency	1628
Common causes of CDC latency	1629
Troubleshooting latency issues	1632
Working with diagnostic support scripts	1647
Oracle support scripts	1649
SQL Server support scripts	1652
MySQL-compatible support scripts	1656
PostgreSQL support scripts	1658
Working with the diagnostic support AMI	1660
Launch a new AWS DMS diagnostic Amazon EC2 instance	1661
Create an IAM role	1661
Run Diagnostic Tests	1662
Next Steps	1666
AMI IDs by region	1666
AWS Systems Patch Manager	1667
Reference	1668
AWS DMS data types	1668
Release notes	1671
AWS DMS 3.6.1 release notes	1672
AWS DMS 3.6.0 release notes	1677
AWS DMS 3.5.4 release notes	1679
AWS DMS 3.5.3 release notes	1687
AWS DMS 3.5.2 release notes	1697
AWS DMS 3.5.1 release notes	1705
Change in Handling of Large Numeric Values for Streaming Targets	1705
AWS DMS 3.5.0 Beta release notes	1721
AWS DMS 3.4.7 release notes	1726
AWS DMS 3.4.6 release notes	1735
AWS DMS 3.4.5 release notes	1741

AWS DMS 3.4.4 release notes	1743
AWS DMS 3.4.3 release notes	1746
AWS DMS 3.4.2 release notes	1748
AWS DMS 3.4.1 release notes	1750
AWS DMS 3.4.0 release notes	1751
AWS DMS 3.3.4 release notes	1753
AWS DMS 3.3.3 release notes	1753
Document history	1755
AWS Glossary	1761

What is AWS Database Migration Service?

AWS Database Migration Service (AWS DMS) is a cloud service that makes it possible to migrate relational databases, data warehouses, NoSQL databases, and other types of data stores. You can use AWS DMS to migrate your data into the AWS Cloud or between combinations of cloud and on-premises setups.

With AWS DMS, you can discover your source data stores, convert your source schemas, and migrate your data.

- To discover your source data infrastructure, you can use DMS Fleet Advisor. This service collects data from your on-premises database and analytic servers, and builds an inventory of servers, databases, and schemas that you can migrate to the AWS Cloud.
- To migrate to a different database engine, you can use DMS Schema Conversion. This service automatically assesses and converts your source schemas to a new target engine. Alternatively, you can download the AWS Schema Conversion Tool (AWS SCT) to your local PC to convert your source schemas.
- After you convert your source schemas and apply the converted code to your target database, you can use AWS DMS to migrate your data. You can perform one-time migrations or replicate ongoing changes to keep sources and targets in sync. Because AWS DMS is a part of the AWS Cloud, you get the cost efficiency, speed to market, security, and flexibility that AWS services offer.

At a basic level, AWS DMS is a server in the AWS Cloud that runs replication software. You create a source and target connection to tell AWS DMS where to extract data from and where to load it. Next, you schedule a task that runs on this server to move your data. AWS DMS creates the tables and associated primary keys if they don't exist on the target. You can create the target tables yourself if you prefer. Or you can use DMS Schema Conversion to create some or all of the target tables, indexes, views, triggers, and so on.

The following diagram illustrates the AWS DMS replication process.

